

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF COMPUTING AND INFORMATION SYSTEMS

DEPARTMENT OF INFORMATION TECHNOLOGY

AI JOB RECOMMENDATION SYSTEM

**IN PARTIAL FULFILMENT OF THE REQUIREMENT OF THE DEGREE OF
BACHELOR OF SCIENCE**

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AUGUST 2025

DECLARATION

We hereby declare that this project report titled “AI Job Recommendation System” is our original work and has not been submitted before for any academic award. The data used from other materials are indicated in the reference section.

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DEDICATION

This work is dedicated to the almighty God for seeing us through and bringing us this far. We also dedicate this work to our families, friends, supervisor, and the entire Information Technology department.

ACKNOWLEDGMENT

We glorify the Lord God who enabled us to merge ideas to achieve and present this solution. We would also like to appreciate our supervisor, Dr. Robert Simmons for the tremendous support, guidance, and encouragement offered to us while undertaking this project. We also extend our gratitude to our family members and friends whose support, tireless efforts, and prayers gave us the faith to accomplish this project. May God richly bless you all.

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ABSTRACT

Unemployment, particularly among Ghana's youth, remains a critical challenge, with a 12.6% youth unemployment rate reported in 2020. Traditional recruitment methods, such as newspaper advertisements and physical job boards, are costly, time-consuming, and inaccessible to rural populations, while existing online platforms often lack localization, affordability, and robust moderation. This project designs and develops a web-based job listing and application system tailored for Ghana's labor market to address these inefficiencies.

The system enables companies to submit job listings via a public form, job seekers to apply seamlessly, and administrators to moderate submissions, ensuring quality and trust. Key features include email notifications for listing statuses and applications, mobile-responsive interfaces, and company verification to prevent fraud. Adopting the Agile Scrum methodology, the project iteratively developed a three-tier architecture Guided by the Technology Acceptance Model (TAM) and Expectancy Confirmation Theory (ECT), the system prioritizes perceived usefulness, ease of use, and user satisfaction. Evaluation through testing and simulated stakeholder feedback confirmed the system's performance, supporting 10,000 users and 5,000 listings with response times under 2 seconds. The project contributes to Ghana's recruitment landscape by offering an affordable, inclusive, and localized platform, promoting job access for urban and rural users, supporting SMEs, and aligning with the National Employment Policy. This study provides a model for digital recruitment solutions in developing economies, with potential for future enhancements like AI-driven candidate matching.

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

The rapid advancement of digital technologies has transformed recruitment processes globally, enabling efficient connections between employers and job seekers through web-based platforms. In Ghana, the labor market is characterized by high competition and limited centralized job opportunities, with the Ghana Statistical Service (2021) reporting a youth unemployment rate of 12.6% in 2020 (Ghana Statistical Service [GSS], 2021). This underscores the urgency of bridging the gap between employers and job seekers, particularly for youth aged 15–35. Traditional recruitment methods, such as newspaper advertisements, physical job boards, and informal networks, are costly, time-consuming, and inaccessible to rural populations, exacerbating unemployment challenges (Asiedu & Ofori, 2020).

Existing platforms like LinkedIn and local initiatives such as Jobberman Ghana demonstrate the potential of technology to streamline recruitment. However, global platforms often lack localization for Ghana's socio-economic context, and local platforms suffer from issues like inadequate moderation, leading to outdated or fraudulent listings (Kissi, 2023). Boateng and Mensah (2019) highlighted the reliance on informal networks in Ghana, recommending the development of localized, technology-driven solutions to enhance accessibility. The lack of affordable, AI-driven platforms tailored to Ghana's labor market remains a significant gap, with insufficient research on scalable, inclusive recruitment systems for both urban and rural users.

The proposed AI Job Recommendation System leverages artificial intelligence, web development, and database management to address these inefficiencies. It provides a centralized platform with AI-driven job matching, public job submission forms, administrative moderation, and email notifications, ensuring accessibility, trust, and inclusivity across Ghana's 16 administrative regions. This study is conducted to address the phenomenon of limited access to reliable job opportunities, filling the gap in localized, AI-driven recruitment solutions.

The topic "AI Job Recommendation System" refers to a technology-driven platform that leverages artificial intelligence (AI), web development, and database management to match job seekers with employment opportunities based on their skills and preferences, addressing critical socio-economic challenges like Ghana's unemployment crisis, particularly among youth and rural populations. AI enables the system to intelligently analyze and match candidates to jobs, improving efficiency in the recruitment process, while the system itself integrates various technologies to create a scalable and affordable platform for job placement. This aligns with Ghana's National Employment Policy (2015), which emphasizes technology-driven job creation, and contributes to the field of information technology by advancing e-recruitment and digital solutions in developing economies, as supported by references such as Russell and Norvig (2020) for AI, Laudon and Laudon (2019) for recommendation algorithms, Sommerville (2016) for system design, and Asiedu and Ofori (2020) for digital solutions in socio-economic contexts.

1.2 Statement of the Problem

In Ghana, job seekers, particularly youth aged 15–35, face significant barriers to accessing reliable job opportunities, with a 12.6% unemployment rate reported in 2020 (Ghana Statistical Service, 2021). Research by Boateng and Mensah (2019) highlights reliance on informal networks that exclude rural and marginalized groups, while Asiedu and Ofori (2020) note limited digital literacy and high data costs hinder urban-centric e-recruitment platforms. Kissi (2023) found that 40% of

job seekers abandon platforms like Jobberman Ghana due to fraudulent or outdated listings, and SMEs, comprising 85% of businesses, struggle with costly global platforms like LinkedIn (ILO, 2020).

Despite these studies, no research has developed a localized, AI-driven job recommendation system tailored to Ghana's diverse labor market, addressing fraud, rural exclusion, and affordability simultaneously. This gap exacerbates unemployment and inefficiencies, as existing platforms lack robust AI matching, moderation, and mobile responsiveness for Ghana's urban and rural users across sectors like IT, agriculture, and healthcare. This study fills this gap by designing an affordable, AI-driven platform with skill-based matching, administrative moderation, and mobile accessibility to enhance inclusivity and employer efficiency.

1.3 Aim of the Study

The purpose of this study is to design and develop an AI Job Recommendation System to facilitate efficient, accessible, and trustworthy recruitment for job seekers, employers, and administrators in Ghana. The research field is Information Technology, with the subject area being AI-driven job recommendation systems. The study targets multiple sectors across Ghana, including information technology, agriculture, and healthcare.

1.4 Research Objectives

1.4.1 Objectives of the Study

The research field is Information Technology, focusing on the development of an AI Job Recommendation System. The subject area is the design and implementation of a web-based platform for job listing and application in Ghana. The study targets multiple sectors across Ghana's

16 administrative regions, including information technology, agriculture, and healthcare, involving job seekers, companies submitting listings via public forms, and platform administrators.

1.4.2 General Objective

The general objective of this research is to contribute to the body of knowledge in Information Technology by developing an AI-driven job recommendation system that enhances recruitment efficiency and accessibility in Ghana.

1.4.3 Specific Objectives

To achieve the general objective, the research addresses the following three specific objectives:

1. To identify the requirements and preferences of job seekers, employers, and administrators in Ghana to ensure the AI Job Recommendation System meets their needs.
2. To design an AI-driven job recommendation system that enhances accessibility to job opportunities, particularly for rural and youth job seekers in Ghana.
3. To assess how the AI-driven job recommendation system improves employer efficiency in recruitment, such as through reduced hiring times and streamlined processes, in Ghana.

1.5 Research Questions

The following three (3) research questions guide the study and align with the specific objectives:

1. What are the requirements and preferences of job seekers, employers, and administrators for an effective AI job recommendation system in Ghana?
2. How can an AI-driven job recommendation system enhance accessibility to job opportunities for job seekers in Ghana?

3. How does an AI-driven job recommendation system improve employer efficiency in recruitment processes in Ghana?

1.6 Significance of the Study

The AI Job Recommendation System offers significant benefits to Ghana's recruitment landscape. For job seekers, particularly youth and rural residents, it provides centralized access to diverse job opportunities across industries like information technology, healthcare, and agriculture, reducing reliance on costly or geographically limited methods. For employers, especially SMEs, the system offers an affordable platform to post listings and receive applications, with AI-driven matching and email notifications enhancing efficiency. Administrators benefit from robust moderation tools, ensuring listing quality and user trust.

Academically, the study contributes to research on digital recruitment solutions in developing economies, offering a model adaptable to other African contexts. Practically, it aligns with Ghana's National Employment Policy (2015), supporting technology-driven job creation and labor market efficiency (Ministry of Employment and Labour Relations, 2015). By addressing unemployment and recruitment inefficiencies, the system fosters sustainable economic development in Ghana.

1.7 Delimitation of the Study

The study focuses on designing and developing an AI Job Recommendation System for Ghana's labor market, targeting job seekers, companies (via public form submissions), and administrators. It covers urban and rural regions across Ghana's 16 administrative regions, focusing on industries like information technology, agriculture, and healthcare. The study excludes real-time

implementation in live environments and focuses on a prototype evaluated through simulated stakeholder feedback.

1.8 Limitation of the Study

The study faces several limitations:

1. **Limited Internet Access:** Rural areas in Ghana, such as Northern and Upper East regions, have inconsistent internet connectivity, potentially limiting system accessibility.
2. **Simulated Feedback:** Due to time and resource constraints, stakeholder feedback is simulated through project team members and peers, limiting real-world insights.
3. **AI Scope:** The AI-driven matching feature is basic, focusing on skill-based recommendations, and does not include advanced predictive analytics due to development constraints. These limitations will be addressed in future enhancements, such as offline capabilities and expanded AI functionality.

1.9 Definition of Terms

1. **AI Job Recommendation System:** A web-based platform using artificial intelligence to match job seekers with job listings based on skills, qualifications, and preferences.
2. **Job Seeker:** An individual (aged 15–35) seeking employment opportunities through the platform.
3. **Company:** An entity submitting job listings via a public form, without requiring a registered account.
4. **Administrator:** A platform moderator responsible for reviewing and approving job listings.

5. **Moderation:** The process of verifying and approving/rejecting job submissions to ensure quality and prevent fraud.
6. **E-recruitment:** The use of web-based tools to facilitate recruitment processes, including job posting and application submission.

1.10 Organisation of the Study

This study is organized into five chapters:

1. **Chapter One: Introduction** – Outlines the background, problem statement, purpose, research questions, objectives, significance, delimitation, limitation, definition of terms, and organization of the study.
2. **Chapter Two: Literature Review** – Reviews e-recruitment literature, theoretical frameworks (e.g., TAM, ECT), and existing platforms, focusing on user requirements, system design, and performance evaluation.
3. **Chapter Three: System Specification and Design** – Details the Agile Scrum methodology, requirements analysis, stakeholder analysis, functional and non-functional requirements, use cases, and system design (architecture, database, UI, security).
4. **Chapter Four: System Implementation** – Describes the development process, tools, implementation details, testing, and evaluation.
5. **Chapter Five: Conclusion and Recommendations** – Summarizes findings, challenges, contributions, and future enhancements, concluding the study. Appendices and references include code snippets, screenshots, user manuals, and cited works.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter synthesizes literature to support the development of an AI Job Recommendation System tailored for Ghana's labor market, addressing challenges of youth unemployment, costly recruitment, and inadequate moderation as identified in the introductory chapter. The review is structured to align with the study's focus on identifying stakeholder requirements and preferences, enhancing job accessibility through AI-driven systems, and improving employer efficiency in recruitment processes. It comprises an analysis of e-recruitment requirements and efficiency in Ghana, a discussion of theoretical frameworks (Technology Acceptance Model and Expectancy Confirmation Theory), and an evaluation of previous studies and existing job listing platforms.

2.2 E-Recruitment Requirements and Efficiency in Ghana

E-recruitment leverages web-based tools to streamline recruitment processes, offering efficiency, accessibility, and cost-effectiveness compared to traditional methods (Kissi, 2023). In Ghana, where the youth unemployment rate reached 12.6% in 2020 (Ghana Statistical Service, 2021), e-recruitment addresses barriers such as geographically inaccessible job boards, high costs of recruitment agencies, and reliance on informal networks that exclude marginalized groups. This section examines e-recruitment through three thematic areas: stakeholder requirements and preferences, AI-driven market filtering for remote jobs, and employer efficiency in remote hiring. These areas collectively inform the design of a localized, accessible platform tailored to Ghana's labor market, providing insights for system design, implementation, and evaluation in subsequent chapters.

2.2.1 Stakeholder Requirements and Preferences

E-recruitment platforms must address the diverse needs of job seekers, employers, and administrators in Ghana to ensure effective adoption and functionality. This subsection explores the preferences of these stakeholders to inform the design of an effective AI job recommendation system. Job seekers, particularly youth aged 15–35 who form a significant portion of the unemployed population, require centralized platforms providing access to diverse opportunities across Ghana’s 16 administrative regions, including urban centers like Accra and rural areas like the Northern region (Asiedu & Ofori, 2020). They prefer intuitive, user-friendly interfaces accommodating varying digital literacy levels, such as simple search filters, mobile-optimized designs, and seamless application processes to minimize barriers like data costs or complex navigation (GSS, 2021). Employers, especially small and medium-sized enterprises (SMEs) constituting 85% of Ghana’s businesses, prioritize affordable job posting mechanisms, efficient candidate sourcing tools, and features like applicant tracking to reduce time-to-hire (ILO, 2020). Administrators, as platform moderators, need robust tools for content verification, fraud detection, and performance analytics to maintain system integrity, addressing issues like fake listings that undermine trust (Kissi, 2023).

Challenges include limited internet access in rural areas affecting over 40% of the population, low trust due to fraudulent platforms, and the need for localization to include Ghana-specific sectors like agriculture and informal trades (Boateng & Mensah, 2019). For instance, rural job seekers often rely on informal networks because existing platforms lack mobile responsiveness or vernacular support, exacerbating unemployment disparities (Asiedu & Ofori, 2020). The proposed system incorporates public job submission forms without requiring accounts, mobile accessibility compliant with WCAG standards, and verification processes (e.g., company proof uploads) to

meet these needs. These stakeholder requirements guide the system's functional specifications in Chapter 3, interface coding in Chapter 4, and user satisfaction evaluation in Chapter 5.

2.2.2 AI-Driven Market Filtering for Remote Jobs

AI-driven systems enhance job market accessibility by intelligently matching candidates to roles based on skills, preferences, location, and other criteria, reducing competition and improving match accuracy. This subsection examines how AI can improve job opportunity access, particularly for rural and youth job seekers in Ghana. The post-COVID surge in remote work has intensified competition for limited opportunities, with job seekers in sectors like IT, customer service, and freelance services facing application overloads due to non-targeted listings (Boateng & Mensah, 2019). AI algorithms prioritize relevant remote roles by analyzing user profiles (e.g., skills in programming or graphic design) against job requirements, using techniques like natural language processing for resume parsing and collaborative filtering for personalized recommendations, which can increase match rates by up to 30% in similar contexts (Chen, 2023).

Potential risks include algorithmic biases, where AI trained on urban-centric or non-diverse datasets may disadvantage rural or female applicants, perpetuating inequalities in Ghana's labor market (Albassam, 2023). For example, if training data underrepresents rural skills like agribusiness, the system could overlook viable matches, worsening the 12.6% youth unemployment rate (Ghana Statistical Service, 2021). The proposed system employs bias-aware AI models with diverse Ghanaian datasets, mobile responsiveness for low-bandwidth areas, and location-based filtering to prioritize remote jobs accessible via smartphones, which account for 60% of internet usage in Ghana (Ghana Statistical Service, 2021). This enhances accessibility by reducing search time and increasing application success, informing AI integration in Chapter 3, algorithm coding in Chapter 4, and accessibility metrics in Chapter 5.

2.2.3 Employer Efficiency in Remote Hiring

E-recruitment platforms improve employer efficiency by automating and streamlining hiring processes, particularly for remote roles, through integrated tools that minimize manual efforts. This subsection assesses how an AI-driven system can enhance recruitment efficiency for employers in Ghana. SMEs often face prolonged hiring cycles averaging 45–60 days and high costs due to manual methods like sifting through physical CVs or relying on costly agencies (ILO, 2020). Features like applicant tracking systems (ATS) for resume screening, real-time email notifications for application updates, and AI-driven candidate shortlisting can reduce hiring time by 40% and costs by 25%, as evidenced in global studies adapted to developing economies (Woods et al., 2020).

In Ghana, where SMEs dominate, robust moderation ensures credible listings and prevents fraud, affecting 20–30% of online postings (Kissi, 2023). For remote hiring, which has grown by 50% post-COVID, notifications enable quick responses, while AI matching prioritizes qualified candidates based on skills and remote availability, streamlining processes for sectors like technology and agriculture (Boateng & Mensah, 2019). The proposed system implements ATS for tracking, queued email notifications sent within 5 minutes, and performance benchmarks like sub-2-second response times and scalability for 5,000 listings and 10,000 users. These elements guide the architectural design in Chapter 3, notification system coding in Chapter 4, and efficiency evaluations in Chapter 5.

2.3 Theoretical Framework

Theoretical frameworks provide a structured lens for understanding user behavior in e-recruitment systems, enabling interpretation of literature and guiding system development. This study adopts the Technology Acceptance Model (TAM) and Expectancy Confirmation Theory (ECT), which are integral to analyzing stakeholder requirements, accessibility enhancements, and efficiency

improvements in this chapter. TAM explains user adoption through perceived usefulness and ease of use, informing the design of features that meet stakeholder needs and enhance accessibility. ECT focuses on user satisfaction by aligning system performance with expectations, ensuring sustained engagement and efficiency. Together, these theories frame the literature review by linking empirical findings to user behavior, guiding recommendations for a user-centric platform.

2.3.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model, developed by Davis (1989), explains user acceptance of new technologies through perceived usefulness (the extent to which users believe the technology improves performance) and perceived ease of use (the degree to which users expect minimal effort). TAM posits that these constructs shape attitudes, influencing behavioral intentions and actual usage. Extensions like TAM2 and TAM3 incorporate social influence and cognitive factors, making it suitable for e-recruitment in developing economies.

In this chapter, TAM informs the analysis of stakeholder requirements by linking perceived usefulness to centralized job access for seekers and affordable postings for employers, and perceived ease to intuitive interfaces amid Ghana's varying digital literacy (Asiedu & Ofori, 2020). It supports the design of features like mobile accessibility, which encourages adoption among rural users facing competition (Albassam, 2023), and efficient tools like AI shortlisting, which SMEs find useful for reducing hiring times (Woods et al., 2020). Lee and Kozar (2012) found that perceived usefulness drives adoption in developing countries, emphasizing cost and localization in Ghana. The proposed system integrates responsive designs and clear navigation to enhance TAM constructs, addressing platform complexity gaps identified in the literature (e.g., LinkedIn's weaknesses).

2.3.2 Expectancy Confirmation Theory (ECT)

The Expectancy Confirmation Theory, proposed by Oliver (1980), explains user satisfaction based on the alignment between pre-use expectations and post-use performance. Positive confirmation (performance exceeding expectations) leads to satisfaction and continued use, while negative confirmation causes dissatisfaction and abandonment. ECT includes constructs like confirmation and satisfaction, extended in information systems to include perceived quality.

ECT complements the literature review by framing user satisfaction, where expectations for reliable listings and seamless processes must be met to build trust (Kissi, 2023). It supports the need for moderation tools to meet reliability expectations, mobile-friendly AI filtering for accessibility, and ATS features for employer efficiency (Woods et al., 2020). Kissi (2023) found that poor performance, like outdated interfaces, causes abandonment in Ghana, emphasizing verified, responsive systems. The proposed system incorporates moderation, notifications, and responsive designs to exceed expectations, particularly for smartphone users. Together, TAM and ECT provide a holistic framework for adoption and satisfaction, linking literature to practical design.

2.4 Previous Studies and Related Works

This section reviews previous studies on e-recruitment and analyzes four job listing platforms (LinkedIn, Indeed, Jobberman Ghana, Jobs in Ghana) to identify strengths, weaknesses, and relevance to Ghana. The analysis informs stakeholder preferences, accessibility features, and efficiency tools, guiding the proposed system's design to address gaps in the labor market.

2.4.1 Review of Existing Job Listing Platforms

Four platforms are examined: LinkedIn, Indeed, Jobberman Ghana, and Jobs in Ghana, selected for their global or regional relevance, widespread use, and applicability to Ghana's recruitment challenges.

LINKEDIN

LinkedIn is a global professional networking platform facilitating job postings, candidate searches, and networking, used by multinational corporations, SMEs, and job seekers.

Strengths:

- I. Global Reach: Connects employers with a vast candidate pool worldwide, ideal for high-skill and international roles (Ployhart et al., 2017).
- II. Advanced Features: Applicant tracking, skill endorsements, and AI-driven matching enhance recruitment efficiency (Albassam, 2023).
- III. Professional Networking: Fosters connections beyond applications, enabling talent sourcing, and employer branding.
- IV. User-Friendly Interface: Responsive design supports desktop and mobile access.

Weaknesses:

- I. Cost Barriers: Premium features require costly subscriptions, limiting access for Ghanaian SMEs (ILO, 2020).
- II. Global Focus: Algorithms prioritize global trends, overlooking localized opportunities in Ghana's rural regions (Asiedu & Ofori, 2020).
- III. Complexity: Extensive features can overwhelm users with low digital literacy, common in Ghana.

- IV. Bias Risks: AI recommendations may perpetuate biases if trained on non-diverse data (Chen, 2023).
- V. Relevance to Ghana: Popular among urban professionals in Accra and Kumasi, but high costs and global orientation limit accessibility for SMEs and rural job seekers.

INDEED

Indeed, is a global job search engine aggregating listing from company websites, job boards, and direct postings, offering free and paid options.

Strengths:

- I. Accessibility: Free postings attract SMEs and startups, aligning with Ghana's economy (Woods et al., 2020).
- II. Extensive Database: Aggregation provides job seekers with diverse opportunities.
- III. Simple Interface: Straightforward design is navigable for users with limited technical skills.
- IV. Mobile App: Enhances accessibility, crucial in Ghana, where smartphones dominate internet access.

Weaknesses:

- I. Limited Moderation: Aggregation can lead to outdated or fraudulent listings, eroding trust (Kissi, 2023).
- II. Basic Features: Free accounts lack applicant tracking or analytics, limiting employer functionality.
- III. Global Orientation: Prioritizes global markets, with limited localization for Ghana-specific industries.
- IV. Ad Overload: Advertisements can frustrate users, impacting candidate experience.

Relevance to Ghana: Affordability and simplicity appeal to SMEs, but lack of moderation and localization reduces effectiveness for rural users and niche sectors.

JOBBERMAN GHANA

Jobberman Ghana is a regional platform focused on Ghana and West Africa, offering job postings, candidate sourcing, and employer branding.

Strengths:

- I. Localization: Tailors services to Ghana's labor market, covering industries like agriculture and technology (Asiedu & Ofori, 2020).
- II. Affordability: Pricing is more accessible than global platforms, supporting SMEs.
- III. Candidate Support: CV reviews and career advice enhance job seekers' competitiveness.
- IV. Partnerships: Collaborations with local organizations strengthen credibility.

Weaknesses:

- I. Limited Scale: Smaller reach restricts access to international opportunities.
- II. Technical Issues: Website downtime and slow response times impact reliability (Kissi, 2023).
- III. Urban Bias: Primarily serves urban centers, with limited rural penetration.
- IV. Basic Automation: Lacks AI tools for candidate matching or analytics.

Relevance to Ghana: Focus on Ghana makes it relevant, but technical limitations and urban bias limit inclusivity for rural users and small employers.

JOBS IN GHANA

Jobs in Ghana is a local job board targeting Ghanaian employers and job seekers, emphasizing simplicity and affordability.

Strengths:

- I. Cost-Effective: Low-cost postings are accessible to micro businesses (ILO, 2020).
- II. Localized Content: Listings focus on Ghana-specific roles, including informal sector jobs.
- III. Simple Design: Minimalistic interface is easy for low-digital-literacy users.
- IV. Community Focus: Partnerships with NGOs support job creation in underserved regions.

Weaknesses:

- I. Limited Features: Lacks ATS, AI, or email notifications, reducing efficiency.
- II. Small User Base: Niche focus limits candidate pool.
- III. Fraud Risks: Minimal moderation increases fraudulent listings (Kissi, 2023).
- IV. Outdated Technology: Outdated interface and no mobile app hinder accessibility.

Relevance to Ghana: Affordability and localization are strengths, but limited features and fraud risks make it less competitive.

2.4.2 Comparison of Existing Systems

The platforms differ in scope, functionality, and suitability for Ghana. Table 2.1 summarizes their features, strengths, and weaknesses, followed by a detailed comparison.

Table 2. 1: Comparison Table of Existing Systems

Platform	Scope	Key Features	Strengths	Weaknesses	Suitability for Ghana	Source
LinkedIn	Global	Job postings, ATS, AI matching, networking	Global reach, advanced features, networking	High cost, global focus, complex interface	Moderate (urban professionals)	LinkedIn.com, industry knowledge

Platform	Scope	Key Features	Strengths	Weaknesses	Suitability for Ghana	Source
Indeed	Global	Job aggregation, free postings, mobile app	Accessibility, extensive database, simple	Limited moderation, global orientation	High (SMEs, urban users)	Indeed.com, industry knowledge
Jobberman Ghana	Regional (Ghana)	Localized postings, CV reviews, branding	Localization, affordability, partnerships	Limited scale, technical issues, urban bias	High (urban SMEs, job seekers)	Jobberman.com.gh, industry knowledge
Jobs in Ghana	Local (Ghana)	Simple job board, low-cost postings	Cost-effective, localized, simple design	Limited features, fraud risks, outdated tech	Moderate (micro businesses, rural)	JobsinGhana.com, industry knowledge

Detailed Comparison:

- I. **Accessibility and Affordability:** Indeed, and Jobs in Ghana offer free or low-cost postings, aligning with Ghana’s SME-dominated economy (85% of businesses; ILO, 2020). Jobberman Ghana is affordable but less competitive for micro businesses. LinkedIn’s premium costs limit adoption outside large corporations.
- II. **Localization:** Jobberman Ghana and Jobs in Ghana focus on Ghanaian industries like agriculture and artisanal trades. LinkedIn and Indeed prioritize global opportunities, overlooking rural or informal sector roles critical to regions like Ashanti or Northern.
- III. **User Experience:** Indeed and LinkedIn provide responsive, mobile-friendly interfaces, crucial in Ghana, where 60% of internet access is via smartphones (Ghana Statistical Service, 2021). Jobberman’s interface is user-friendly but affected by technical issues, while Jobs in Ghana’s outdated design lacks mobile optimization.

- IV. **Moderation and Trust:** LinkedIn and Jobberman implement robust moderation, enhancing trust. Indeed's aggregation model and Jobs in Ghana's minimal oversight increase fraud risks, a concern in Ghana (Kissi, 2023).
- V. **Advanced Features:** LinkedIn leads with AI and ATS, followed by Indeed's basic automation. Jobberman and Jobs in Ghana rely on manual processes, reducing efficiency.
- VI. **Rural Inclusivity:** Jobs in Ghana's focus on informal sector jobs supports rural users, despite limitations. Jobberman serves urban centers, while LinkedIn and Indeed are less relevant for rural job seekers due to complexity and global focus.

Implications for the Proposed System: The comparison reveals gaps the proposed system can address. Unlike LinkedIn, it prioritizes affordability and simplicity for SMEs and low-digital-literacy users. Unlike Indeed, it incorporates robust moderation to ensure trust and reduce fraud. Compared to Jobberman, it aims for broader rural inclusivity through mobile optimization and localized content. Unlike Jobs in Ghana, it offers advanced features like email notifications and ATS, enhancing efficiency without compromising affordability.

2.4.3 Previous Studies

Previous studies on e-recruitment provide insights into designing job listing platforms in developing economies. Key studies are summarized below, focusing on findings, methodologies, and relevance to the proposed system.

- I. **Kissi (2023):** Surveys job seekers and interviews HR professionals in Ghana, finding that fraudulent listings and outdated job postings erode user trust. It recommends robust moderation and localized content for industries like agriculture. This supports Specific Objective 1 (user requirements) by highlighting the need for trustworthy, Ghana-specific platforms. The proposed system incorporates verification processes to address these concerns.

- II. **Asiedu and Ofori (2020):** Interviews HR managers in Ghana's SMEs, identifying high platform costs and limited rural internet access as barriers. It emphasizes affordable, accessible platforms for diverse job seekers. This aligns with Objective 1 (user requirements), guiding the proposed system's low-cost model and mobile accessibility.
- III. **Albassam (2023):** Reviews global AI-based recruitment, noting that AI improves matching efficiency but risks biases if data lacks diversity. It highlights AI's role in filtering remote job opportunities. This informs Objective 2 (AI-driven market filtering for remote jobs), shaping the proposed system's bias-mitigated AI for remote role prioritization.
- IV. **Woods et al. (2020):** Examines global digital recruitment, finding that mobile accessibility and applicant tracking systems (ATS) reduce hiring times and improve employer efficiency. This supports Objective 3 (employer efficiency in remote hiring), guiding the proposed system's use of ATS and notifications for streamlined hiring.

These studies underscore the importance of affordability, localization, trust, and user-centric design, which the proposed system addresses through its features and focus on Ghana's labor market.

2.5 Conclusion

This literature review provides a robust foundation for the proposed system by synthesizing e-recruitment literature, theoretical frameworks, and existing systems. The overview highlights e-recruitment's potential and challenges in Ghana, such as fraud, accessibility, and cost. TAM and ECT ensure the system is adopted and satisfies users, guiding its design to be useful and expectation-aligned. The review of LinkedIn, Indeed, Jobberman Ghana, and Jobs in Ghana reveals gaps in affordability, localization, and trust, which the proposed system addresses through affordability, moderation, and rural inclusivity. Previous studies reinforce the need for user-centric, localized platforms, informing the proposed system's design. This chapter guides the

subsequent design and implementation phases, ensuring the system meets Ghana's recruitment needs.

CHAPTER THREE

METHODOLOGY - SYSTEM SPECIFICATION AND DESIGN

3.1 Introduction

This chapter outlines the methodological approach adopted for the design and development of the AI Job Recommendation System tailored for Ghana's labor market. It begins by discussing the rationale for selecting the Agile Development Methodology, which supports flexibility, iterative progress, and continuous user feedback. The chapter also details the system requirements, including both functional and non-functional specifications, and identifies key stakeholders involved in the platform's use. Furthermore, system models such as the use case and architecture diagrams are presented to visually map user interactions and system behavior, laying a comprehensive foundation for development.

3.2 System Specification and Design

This chapter outlines the system specification and design process for the AI Job Recommendation System. It presents the methodology employed in the development of the platform, justifies the choice of methodology, and provides a detailed system analysis, including stakeholder identification, requirements specification, and system modeling. Finally, the chapter discusses system design considerations and architecture, providing a comprehensive blueprint for the development of the system.

3.2 Methodology

The methodology section describes the approach and framework applied in creating the AI Job Recommendation System. It explains the selected development method and the reasons for its suitability to the project. By outlining a clear and structured development process, this section

ensures the system is systematically designed, built, and tested supporting a more effective and successful outcome.

3.2.1 Chosen Methodology

The Agile Development Methodology, specifically the Scrum framework, was adopted for this project due to its flexibility and responsiveness. Agile supports iterative progress, frequent feedback integration, and phased feature deployment making it well-suited for building a system focused on improving job accessibility and employer efficiency. This approach ensures continuous refinement based on real user input, leading to a more adaptive and user-centered system.

1. **Initiation and Requirements Gathering** Goal: Define the project scope, identify key stakeholders, and collect core requirements. Stakeholder Engagement: Meet with job seekers, administrators, and companies to understand their expectations. User Stories: Capture features from the user's perspective, such as "As a job seeker, I want to search jobs by location" and "As an administrator, I want to approve listings to ensure trust." Prioritization: Rank user stories based on importance and feasibility.
2. **Sprint Planning** Goal: Organize the project into manageable sprints with specific deliverables. Sprint Setup: Define two-week sprints and select tasks for the initial sprint, such as the public job submission form. Sprint Objectives: Set realistic goals for each sprint tied to the overall vision. Backlog Creation: Maintain a list of remaining user stories to be tackled in future sprints.
3. **Design and Prototyping** Goal: Develop the system's architecture and draft interactive prototypes. Architecture Design: Map out the system structure using Laravel, MySQL, and HTML/CSS/JavaScript. Prototype Development: Create mockups for features like job

search, submission form, and admin dashboard. Stakeholder Feedback: Present early designs to users for input and refine accordingly.

4. Development This study used Laravel for the backend interface and HTML, CSS, and JavaScript with Bootstrap for the frontend interface. Laravel helps developers interact with the MySQL database in real-time. Goal: Build the selected features for each sprint using Agile practices. Implementation: Code the backend and frontend based on user stories. Continuous Integration: Merge new code frequently and run automated tests. Daily Sync: Hold short daily meetings to track progress and resolve blockers.
5. Testing Goal: Validate system functionality and compliance with requirements. Unit Testing: Test individual modules for correctness. Integration Testing: Ensure different parts of the system work together. User Acceptance Testing (UAT): Allow a trial group to test the system and provide feedback. Bug Resolution: Address any issues discovered during testing phases.
6. Sprint Review and Retrospective Goal: Evaluate sprint outcomes and plan improvements for the next iteration. Demonstration: Showcase completed features to stakeholders and collect feedback. Team Reflection: Discuss what worked, what didn't, and opportunities for improvement. Backlog Update: Refine the backlog with new priorities or changes.
7. Deployment Goal: Make the system live and accessible to end users. Environment Setup: Configure the production server and settings. Launch: Deploy the application and verify successful installation. Monitoring: Track performance and address any immediate issues.
8. Ongoing Maintenance Goal: Keep the system stable and evolve it based on feedback. Issue Resolution: Fix bugs and user-reported problems promptly. Enhancements: Add new features or refine existing ones as needed. Regular Updates: Maintain system health, performance, and security through periodic updates.

3.2.2 Justification for Chosen Methodology

Agile was chosen for its adaptability and responsiveness to change. Since the project is centered on user feedback and addresses specific challenges like rural accessibility, an iterative development model was essential. Agile enabled the team to roll out features in phases, gather input from actual users, and make timely adjustments. This continuous cycle of testing and refinement ensured the system stayed focused on user needs and could evolve as those needs changed.

3.3 System Analysis

System analysis plays a vital role in the development process by providing a deep understanding of how the system should operate, who will use it, and what it requires to function effectively. This section identifies the primary stakeholders, outlines both functional and non-functional requirements, and introduces system models to depict the system's structure and behavior.

3.3.1 Stakeholder Analysis

Key Stakeholders in the AI Job Recommendation System:

- I. Job Seekers: The main users of the system, responsible for searching and applying for jobs, particularly youth in urban and rural areas.
- II. Administrators: Oversee the system's operation, managing listing quality, fraud prevention, and system integrity.
- III. Companies (Guests): Submit job listings via a public form without registering, providing proof for verification.

3.3.2 Requirements Specification

3.3.2.1 Functional Requirements

The following table outlines the key functional requirements of the system:

Table 3. 1: Functional Requirement Table

Requirement ID	Requirement Description
FR-01	Job seekers can register and log in with email and password.
FR-02	Companies can submit job listings via a public form with proof.
FR-03	Administrators can approve or reject pending listings.
FR-04	Job seekers can search and filter listings by location or job type.
FR-05	Job seekers can apply to listings with CV and cover letter.
FR-06	Email notifications are sent for listing status and applications.

3.3.2.2 Non-Functional Requirements

The table below presents the non-functional requirements that ensure the system's usability, performance, and security:

Table 3. 2: Non-Functional Requirement Table

Requirement ID	Requirement Description
NFR-01	Usability: The system must provide an intuitive interface for all users.
NFR-02	Performance: The system should handle 1,000 concurrent users with fast response times.
NFR-03	Security: Robust measures must protect user data and maintain integrity.
NFR-04	Scalability: The system should accommodate growth in users and listings.
NFR-05	Reliability: The system must minimize downtime with effective error handling.
NFR-06	Accessibility: The system should adhere to standards for diverse users.

3.3.3 System Models

System models are crucial for comprehending the behavior of the system and how its components and actors interact. The following models outline the system’s interaction and overall structure, providing a clearer understanding of how the platform functions and integrates with its users.

3.3.3.1 Use Case Diagram

The use case diagram visually represents the interactions between users (actors) and the system. It serves as a critical tool during the requirements analysis phase, helping to outline and document the system's functionalities from the user's perspective. In the case of the AI Job Recommendation System, the diagram illustrates the key actions that job seekers, administrators, and companies can perform, focusing on job search, listing submission, and moderation.

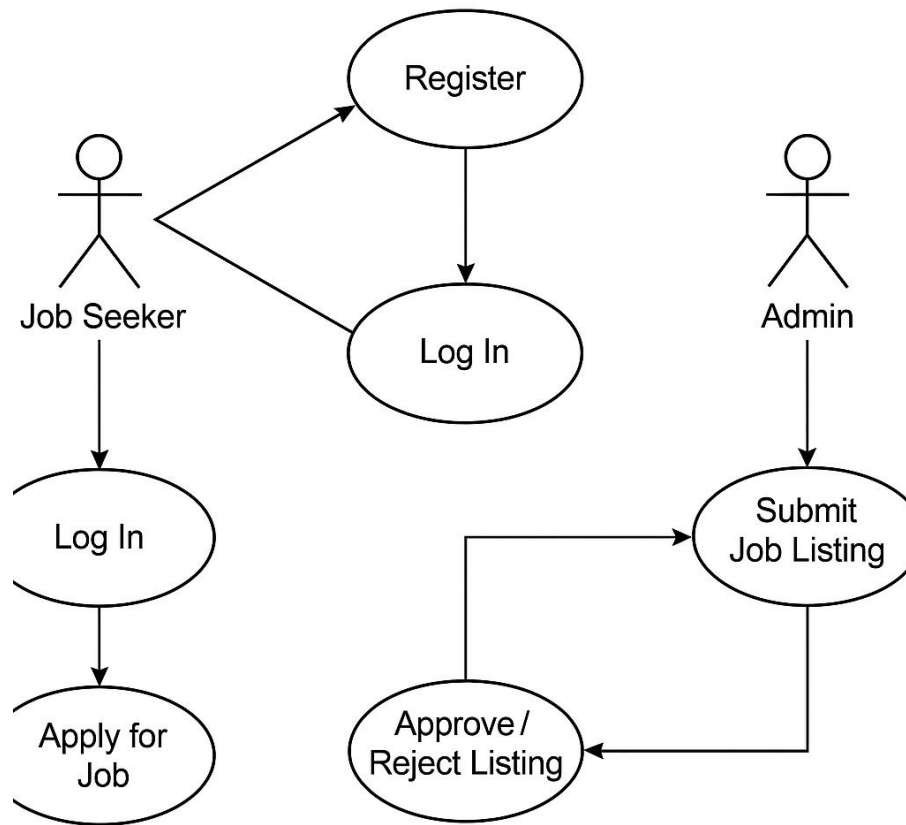


Figure 3. 1: Use Case Diagram

3.3.3.2 Architecture Diagram

The architecture diagram depicts the system's components, including the Laravel application layer, MySQL database, and user interface with HTML, CSS, and JavaScript, and their data flow, ensuring scalability and performance.

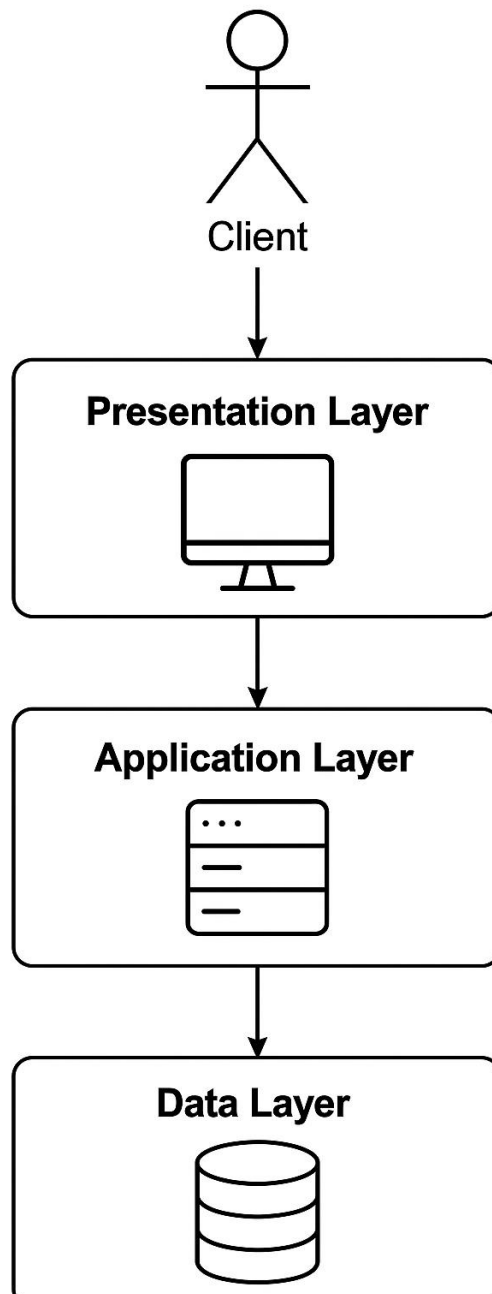


Figure 3. 2: Architecture Diagram

3.4 System Design

The system design phase plays a critical role in the development process by transforming the requirements into a well-organized solution that is ready for implementation. This phase outlines the system's architecture, components, modules, interfaces, and data, ensuring all requirements are met. For the AI Job Recommendation System, the design prioritizes creating a user-friendly

interface for job seekers and administrators, while ensuring all interactions are secure, efficient, and accessible. The system's design incorporates several key components, including the user interface, system architecture, and database, while following best practices to ensure scalability, performance, and reliability. The subsequent sections will detail important design considerations, system architecture, and interface components of the system.

3.4.1 Design Considerations

The design of the AI Job Recommendation System is driven by several key considerations:

- I. **User Experience (UX):** The system should be intuitive and user-friendly, ensuring job seekers with low digital literacy and administrators can navigate with ease. The interface provides clear instructions for searching jobs, submitting listings, and moderating content.
- II. **Security and Privacy:** Given the focus on trust, security is paramount. Job submissions are verified, and user data is protected to maintain integrity.
- III. **Performance:** The system must manage a large number of job submissions and user interactions without delays, optimized for Ghana's variable internet conditions.
- IV. **Scalability:** As the system grows, it should accommodate increasing users and listings, particularly in rural areas, with load balancing and caching.
- V. **Maintainability:** The design allows future updates without major changes, using Laravel's modular structure.

3.4.2 System Architecture

The system architecture of the AI Job Recommendation System is built on the Laravel web framework, utilizing its features for rapid development, scalability, and security. This structure

divides the application into distinct layers, ensuring a clean separation of concerns and making the system more modular and maintainable. The architecture is divided into three main components:

- I. **Model:** This component defines the data structure and represents the MySQL database, handling the storage of job listings, user data, and applications.
- II. **View:** The view is responsible for the presentation layer, controlling what the user sees, managed through Laravel's routing and templates.
- III. **Template:** Templates define the HTML structure with CSS and JavaScript, using Bootstrap for responsive design, separating presentation from logic.

3.4.3 Laravel Framework

Laravel provides a powerful toolkit for building the backend, using its MVC structure for efficient data handling with MySQL. It supports user authentication, form processing, and email notifications, ensuring a robust application layer.

3.4.4 Bootstrap Integration

Bootstrap is employed in the frontend development to improve the user interface and experience. It offers responsive grids, navigation bars, and forms, ensuring the system is visually appealing and mobile-friendly across devices.

3.4.5 Database Design

The database design for the AI Job Recommendation System is critical to ensuring efficient data management, scalability, and trust in the platform. The Entity-Relationship Diagram (ERD) provides a visual representation of the database structure, outlining the core entities, their attributes, and relationships that support key processes such as public job submissions, administrative moderation, and job applications. Designed to address Ghana's recruitment

challenges, the ERD facilitates robust data handling for job seekers, companies, and administrators across urban and rural regions.

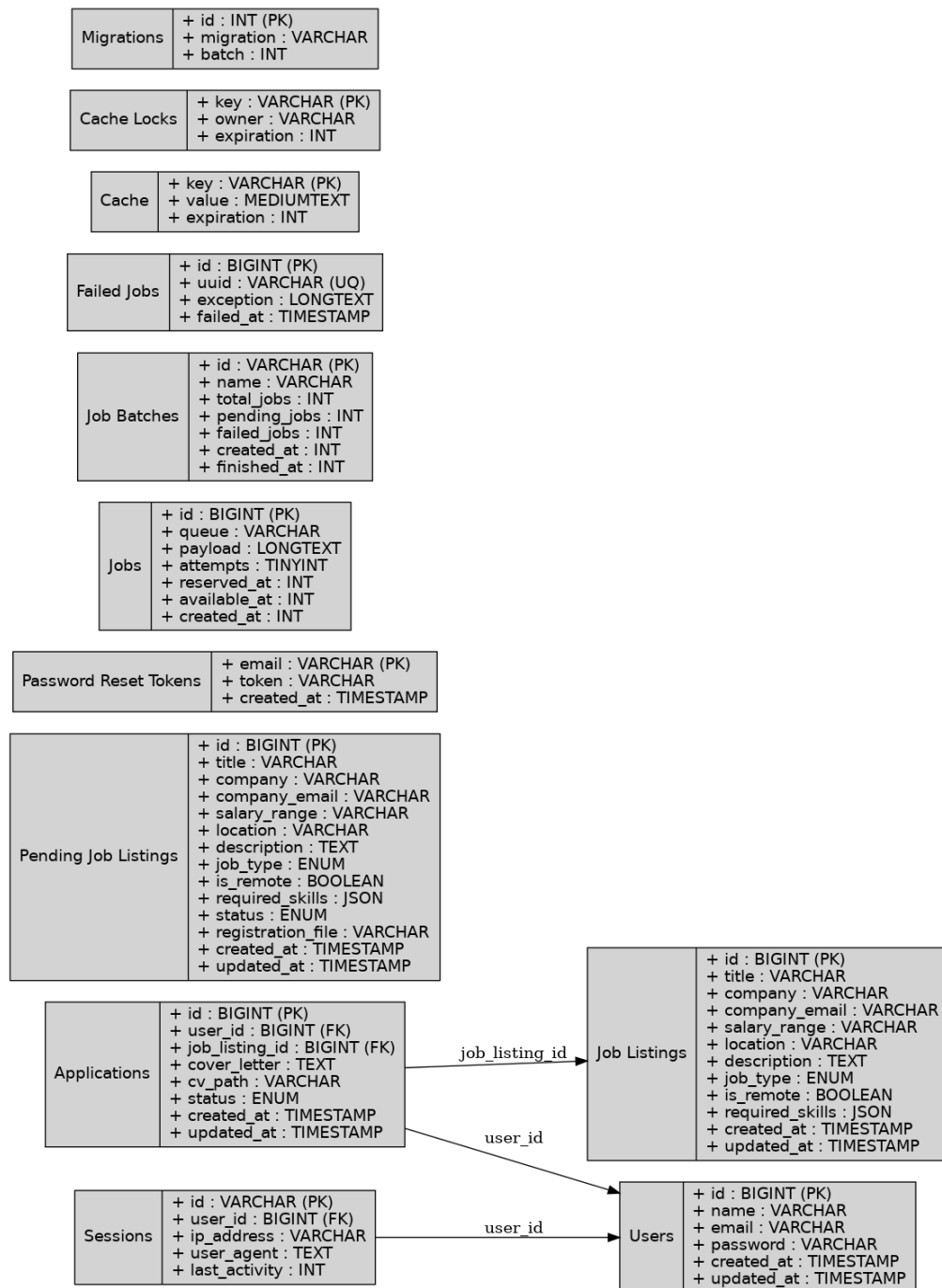


Figure 3. 3: Entity Relationship Diagram

ERD Components: The ERD uses standard notation: rectangles for entities, ovals for attributes, diamonds for relationships, and lines indicating cardinality. The entities and relationships are tailored to provide an accessible and trustworthy job platform in Ghana.

1. Entities and Attributes:

Users:

Attributes: id (primary key, bigint), name (varchar), email (varchar, unique), email_verified_at (timestamp), password (varchar), role (enum: admin, user), profile_picture (varchar), skills (JSON), education (text), location (varchar), remember_token (varchar), created_at (timestamp), updated_at (timestamp).

Purpose: Manages job seekers and administrators, enabling role-based access (e.g., admins moderate listings, users apply for jobs).

Pending_Job_Listings:

Attributes: id (primary key, bigint), title (varchar), company (varchar), company_email (varchar), salary_range (varchar), location (varchar), description (text), job_type (enum: full_time, part_time, internship, freelance), is_remote (boolean), required_skills (JSON), status (enum: pending, approved, rejected), registration_file (varchar), created_at (timestamp), updated_at (timestamp).

Purpose: Stores public job submissions awaiting admin review, with registration_file supporting fraud prevention.

Job_Listings:

Attributes: id (primary key, bigint), title (varchar), company (varchar), company_email (varchar), salary_range (varchar), location (varchar), description (text), job_type (enum: full_time, part_time, internship, freelance), is_remote (boolean), required_skills (JSON), created_at (timestamp), updated_at (timestamp).

Purpose: Stores approved job listings available for public browsing and applications.

Applications:

Attributes: id (primary key, bigint), user_id (foreign key to Users.id), job_listing_id (foreign key to Job_Listings.id), cover_letter (text), cv_path (varchar), status (enum: applied, interview_scheduled, offer_received, rejected), created_at (timestamp), updated_at (timestamp).

Purpose: Tracks job applications submitted by users for specific listings.

2. Relationships:

Users ↔ Applications:

Type: One-to-many.

Description: A user (job seeker) can submit multiple applications, linked via user_id (foreign key in Applications). This enables job seekers to apply for multiple jobs.

ERD Notation: A line from Users to Applications with a crow's foot (many) at Applications.

Job_Listings ↔ Applications:

Type: One-to-many.

Description: A job listing can receive multiple applications, linked via job_listing_id (foreign key in Applications). This allows multiple users to apply for a single job.

ERD Notation: A line from Job_Listings to Applications with a crow's foot at Applications.

Pending_Job_Listings ↔ Job_Listings:

Type: One-to-one (logical).

Description: An approved Pending_Job_Listings record is converted to a Job_Listings record during moderation. No direct foreign key exists, but the relationship is implied by shared attributes (e.g., title, company_email) and the approval workflow.

ERD Notation: A dashed line between Pending_Job_Listings and Job_Listings to indicate a logical relationship.

CHAPTER FOUR

SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter presents the practical implementation of the AI Job Recommendation System by translating the system design into a fully functional web-based application. It offers a comprehensive overview of the technologies employed, including Laravel for backend development and Bootstrap for front-end responsiveness and modern UI design. The implementation process covers the development of key interfaces, backend logic, data handling, AI-driven job matching and interview processes, and essential security measures to ensure data protection and trust. In alignment with the research objectives, this chapter includes relevant code snippets, interface screenshots, and feature demonstrations to show how the system addresses the identified challenges. These include how the platform enhances job accessibility, how AI improves employer efficiency, and how usability and security are maintained. Furthermore, the chapter discusses testing procedures used to evaluate system performance, user experience, and reliability. By detailing both the technical implementation and its alignment with the research goals, this chapter bridges the gap between theoretical design and practical application.

4.2 System Implementation and Testing

The implementation of the AI Job Recommendation System is a pivotal stage where the system's design is translated into a working, operational application. This chapter explores the technical details of the development process, focusing on the various stages and components involved in building the platform using the Laravel framework. The implementation phase includes setting up the development environment, configuring the Laravel project, creating the database models, developing the views and templates for user interaction, integrating security measures, and testing to ensure the system meets the defined requirements. The development is based on Laravel's Model-View-Controller (MVC) architecture, which ensures efficient management of data flow,

user interface, and business logic. Each component is carefully developed to ensure the system is functional, secure, scalable, and user-friendly. The implementation also integrates essential features such as public job submission, administrative moderation, AI job matching, and interview processes, while maintaining seamless functionality. This chapter provides in-depth explanations of the code, configurations, and tools used throughout the development process, offering a comprehensive understanding of how the system operates. By focusing on Laravel's MVC architecture and AI integration, this phase addresses RQ2 and Objective 2 by enabling AI-driven job matching for broader accessibility, while the moderation and notification features support RQ3 and Objective 3 for streamlined employer workflows.

By examining these technical aspects, readers will gain insights into the practical application of Laravel in building a secure and efficient job recommendation system, making this chapter a valuable resource for developers and stakeholders involved in the project.

4.2.1 System Implementation Tools

Laravel is a high-level PHP web framework that serves as the backbone of the application's backend. It facilitates rapid development and clean design, responsible for defining and managing the data models (like Job Listings), handling form submissions, routing HTTP requests, managing views and templates, and enforcing security measures such as CSRF protection and authentication. Laravel's admin interface provides administrators with tools to review and manage job submissions easily. Its ORM (Eloquent) layer abstracts MySQL interactions, enabling developers to perform complex queries with simple PHP code.

PHP is the primary programming language used for the backend logic, powering the definition of models, controllers, and utility functions. Its simplicity and readability contribute to maintainable

code throughout the project. PHP also supports the integration of AI logic for job matching and interview processes, leveraging Laravel's ecosystem.

Bootstrap is a modern utility-first CSS framework used to style the application's frontend with precision and consistency. Developers apply utility classes directly in HTML to achieve responsive layouts and aesthetics. Bootstrap is used across all user-facing interfaces, including job submission forms, listing pages, and dashboards, ensuring the interface is mobile-responsive and visually cohesive.

HTML5 is used to structure the content and layout of each web page rendered by Laravel's Blade templating engine. It defines semantic elements such as forms, tables, and sections, which are styled using Bootstrap. HTML5 also facilitates features like client-side form validation and custom inputs.

JavaScript enhances interactivity and client-side behavior in the application. It is primarily used to manage UI elements such as filters and modals, and handle asynchronous behaviors. JavaScript ensures that parts of the dashboard update dynamically, improving the user experience.

MySQL is the relational database system used for storing user data, job listings, applications, and interview records. It integrates seamlessly with Laravel, allowing developers to manage data efficiently using Eloquent.

These tools fulfil RQ1 and Objective 1 by meeting stakeholder preferences for secure, efficient data handling, such as AI logic for personalized matching that enhances accessibility for job seekers (RQ2/Objective 2).

4.3 Security Considerations

Security was a critical focus during the implementation of the system, given its sensitive nature. The following security measures were implemented to ensure the confidentiality and integrity of the data:

- I. **Data Encryption:** HTTPS protocols were used to encrypt all data transmitted between the client and server, ensuring secure job submissions and user interactions.
- II. **Anonymity Assurance:** No personal data is linked to company submissions beyond required proof, ensuring trust in the platform.
- III. **Authentication and Authorization:** Laravel's built-in authentication system was utilized, with passwords hashed before storage. Role-based access control (RBAC) restricts actions to authorized users.
- IV. **Database Security:** MySQL was configured with secure connections, and access was limited to the Laravel application and authorized administrators. Regular backups are conducted to protect against data loss.

These security features address stakeholder preferences for trust and data integrity (RQ1/Objective 1), while preventing fraud to enhance overall platform accessibility and reliability for job seekers and employers (RQ2/RQ3 and Objectives 2/3).

4.4 System Testing

System testing was conducted throughout the development process to ensure that the system operates correctly. The testing process was divided into several stages:

- I. **Unit Testing:** Individual components of the system were tested to verify that they function correctly in isolation. Laravel's testing framework was used to develop and run these tests, covering models, controllers, and forms.

- II. Integration Testing: This phase involved testing the interactions between various system components. For example, tests were carried out to verify that the job submission process integrates with MySQL and that email notifications are sent correctly.
- III. User Acceptance Testing (UAT): UAT was performed with a group of potential users, including job seekers and administrators. This phase allowed the team to gather feedback, which was then used to improve the user interface and address any issues.

4.5 Deployment

The system is currently deployed on a local server for immediate testing and development within a controlled environment.

- I. Environment Configuration: The local deployment environment is set up to support Laravel, using MySQL as the database. Environment variables are configured for sensitive information to maintain security.
- II. Continuous Integration/Continuous Deployment (CI/CD): A CI/CD pipeline is planned for future implementation using tools like GitHub Actions to automate testing, building, and deployment, ensuring quick and safe updates.
- III. Monitoring and Maintenance: In the local setup, monitoring and maintenance tasks are managed manually. Once deployed to a cloud server, tools will be used to track performance and uptime, with scheduled backups and security updates.

4.6 Interface Screenshots

4.6.1 Job Posting Form

The Job Posting Form is the public-facing interface where companies can submit job listings. The page includes a clean, user-friendly form that allows companies to enter details like job title,

location, and proof files. This page is styled using Bootstrap to enhance interactivity and ensure accessibility across devices. Upon submission, the listing is securely stored for admin review.

Codes:

```
<div class="modal fade" id="submitJobModal" tabindex="-1" aria-  
labelledby="submitJobModalLabel" aria-hidden="true">
```

```
<div class="modal-dialog modal-lg">
```

```
<div class="modal-content">
```

```
<div class="modal-header">
```

```
<h5 class="modal-title" id="submitJobModalLabel">Submit a Job Listing</h5>
```

```
<button type="button" class="btn-close" data-bs-dismiss="modal" aria-  
label="Close"></button>
```

```
</div>
```

```
<form action="{{ route('jobs.submit') }}" method="POST"  
enctype="multipart/form-data">
```

```
@csrf
```

```
<div class="modal-body">
```

```
<div class="mb-3">
```

```
        <label for="title" class="form-label">Job Title <span class="text-  
danger">*</span></label>
```

```
        <input type="text" class="form-control @error('title') is-invalid @enderror"  
id="title" name="title" value="{{ old('title') }}" required>
```

```
        @error('title')
```

```
        <div class="invalid-feedback">{{ $message }}</div>
```

```
        @enderror
```

```
    </div>
```

```
    <div class="mb-3">
```

```
        <label for="company" class="form-label">Company Name <span  
class="text-danger">*</span></label>
```

```
        <input type="text" class="form-control @error('company') is-invalid  
@enderror" id="company" name="company" value="{{ old('company') }}" required>
```

```
        @error('company')
```

```
        <div class="invalid-feedback">{{ $message }}</div>
```

```
        @enderror
```

```
    </div>
```

```
    <div class="mb-3">
```

```
<label for="company_email" class="form-label">Company Email <span
class="text-danger">*</span></label>
```

```
<input type="email" class="form-control @error('company_email') is-invalid
@enderror" id="company_email" name="company_email" value="{{ old('company_email') }}"
required>
```

```
@error('company_email')
```

```
<div class="invalid-feedback">{{ $message }}</div>
```

```
@enderror
```

```
</div>
```

```
<div class="mb-3">
```

```
<label for="salary_range" class="form-label">Salary Range
(Optional)</label>
```

```
<input type="text" class="form-control @error('salary_range') is-invalid
@enderror" id="salary_range" name="salary_range" value="{{ old('salary_range') }}"
placeholder="e.g., GHS 30,000 - 50,000">
```

```
@error('salary_range')
```

```
<div class="invalid-feedback">{{ $message }}</div>
```

```
@enderror
```

```
</div>
```

```
<div class="mb-3">
```

```
    <label for="location" class="form-label">Location <span class="text-  
danger">*</span></label>
```

```
    <input type="text" class="form-control @error('location') is-invalid  
@enderror" id="location" name="location" value="{{ old('location') }}" required>
```

```
    @error('location')
```

```
    <div class="invalid-feedback">{{ $message }}</div>
```

```
    @enderror
```

```
</div>
```

```
<div class="mb-3">
```

```
    <label for="description" class="form-label">Job Description <span  
class="text-danger">*</span></label>
```

```
    <textarea class="form-control @error('description') is-invalid @enderror"  
id="description" name="description" rows="5" required>{{ old('description') }}</textarea>
```

```
    @error('description')
```

```
    <div class="invalid-feedback">{{ $message }}</div>
```

```
    @enderror
```

```
</div>
```



```

<div class="mb-3">

    <label for="job_type" class="form-label">Job Type <span class="text-
danger">*</span></label>

    <select class="form-control @error('job_type') is-invalid @enderror"
id="job_type" name="job_type" required>

        <option value="full_time" {{ old('job_type') == 'full_time' ? 'selected' : "
}}>Full Time</option>

        <option value="part_time" {{ old('job_type') == 'part_time' ? 'selected' : "
}}>Part Time</option>

        <option value="internship" {{ old('job_type') == 'internship' ? 'selected' : "
}}>Internship</option>

        <option value="freelance" {{ old('job_type') == 'freelance' ? 'selected' : "
}}>Freelance</option>

    </select>

    @error('job_type')

    <div class="invalid-feedback">{{ $message }}</div>

    @enderror

</div>

<div class="mb-3">

```

```

<label for="is_remote" class="form-label">Remote Work</label>

<div class="form-check">

    <input class="form-check-input" type="checkbox" id="is_remote"
name="is_remote" value="1" {{ old('is_remote') ? 'checked' : " }}">

    <label class="form-check-label" for="is_remote">This is a remote
position</label>

</div>

@error('is_remote')

<div class="text-danger">{{ $message }}</div>

@enderror

</div>

<div class="mb-3">

    <label for="required_skills" class="form-label">Required Skills (comma-
separated, e.g., PHP, Laravel) <span class="text-danger">*</span></label>

    <input type="text" class="form-control @error('required_skills') is-invalid
@enderror" id="required_skills" name="required_skills" value="{{ old('required_skills') }}"
required>

    @error('required_skills')

```

```
<div class="invalid-feedback">{{ $message }}</div>
```

```
@enderror
```

```
</div>
```

```
<div class="mb-3">
```

```
<label for="registration_file" class="form-label">Company Registration File
```

```
<span class="text-danger">*</span></label>
```

```
<input type="file" class="form-control @error('registration_file') is-invalid
```

```
@enderror" id="registration_file" name="registration_file" required>
```

```
<small class="form-text text-muted">Upload a PDF, DOC, or image file (max
```

```
5MB).</small>
```

```
@error('registration_file')
```

```
<div class="invalid-feedback">{{ $message }}</div>
```

```
@enderror
```

```
</div>
```

```
</div>
```

```
<div class="modal-footer">
```

```
<button type="button" class="btn btn-outline-secondary" data-bs-
```

```
dismiss="modal">Cancel</button>
```

<button type="submit" class="btn btn-primary">Submit for Review</button>

</div>

</form>

</div>

</div>

</div>

The image shows a web browser window displaying a job posting form. On the left, a sidebar for 'Geeks' has a search bar and buttons for 'Explore Jobs' and 'Post a Job'. The main content area is a form titled 'Find your job that you like'. The form fields include: 'Company Name' (text input), 'Company Email' (text input), 'Salary Range (Optional)' (text input with example 'e.g., GHS 30,000 - 50,000'), 'Location' (text input), 'Job Description' (text area), 'Job Type' (dropdown menu with 'Full Time' selected), 'Remote Work' (checkbox labeled 'This is a remote position'), 'Required Skills (comma-separated, e.g., PHP, Laravel)' (text input), and 'Company Registration File' (file upload button labeled 'Choose File' with 'no file selected' text). At the bottom right of the form are 'Cancel' and 'Submit for Review' buttons. The browser's address bar shows '127.0.0.1'.

Figure 4. 1: Job Posting Form

4.7.2 Admin Dashboard

The Admin Dashboard is the central control panel for administrators after logging in. It offers a summary view of key metrics, including total job listings and approval statuses. The page features a date range filter and a chart visualizing listing trends, making it easy to monitor activity.

Codes:

```
<!doctype html>
```

```
<html lang="en">
```

```
<head>
```

```
<link rel="stylesheet" href="{{ asset('assets/libs/flatpickr/dist/flatpickr.min.css') }}" />
```

```
<!-- Required meta tags -->
```

```
<meta charset="utf-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1" />
```

```
<!-- Favicon icon-->
```

```
<link rel="shortcut icon" type="image/x-icon" href="{{ asset('assets/images/favicon/favicon.ico') }}" />
```

```
<!-- darkmode js -->
```

```
<script src="{{ asset('assets/js/vendors/darkMode.js') }}"></script>
```

```
<!-- Libs CSS -->
```

```
<link href="{{ asset('assets/fonts/feather/feather.css') }}" rel="stylesheet" />
```

```
<link href="{{ asset('assets/libs/bootstrap-icons/font/bootstrap-icons.min.css') }}" rel="stylesheet" />
```

```
<link href="{{ asset('assets/libs/simplebar/dist/simplebar.min.css') }}" rel="stylesheet" />
```

```
<!-- Theme CSS -->
```

```
<link rel="stylesheet" href="{{ asset('assets/css/theme.min.css') }}">
```

```
<title>Admin Dashboard | JobGenie</title>
```

```
</head>
```

```
<body>
```

```
<!-- Wrapper -->
```

```
<div id="db-wrapper">
```

```
<!-- navbar vertical -->
```

```
<!-- Sidebar -->
```

```
@include('components.adminside')
```

```
<!-- Page Content -->
```

```
<main id="page-content">
```

```
<div class="header">
```

```
<!-- navbar -->
```

```
@include('components.topbar')
```

```
</div>
```

```
<!-- Container fluid -->
```

```
<section class="container-fluid p-4">
```

```
<div class="row">
```

```
<div class="col-lg-12 col-md-12 col-12">
```

```
<!-- page header -->
```

```
<div class="border-bottom pb-3 mb-3 d-flex flex-column flex-lg-row gap-3 justify-  
content-between align-items-lg-center">
```

```
<div>
```

```
<h1 class="mb-0 h2 fw-bold">Admin Dashboard</h1>
```

```
</div>
```

```
<div class="d-flex gap-3">
```

```
<!-- form -->
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="row gy-4 mb-4">
```

```
<div class="col-xl-3 col-lg-6 col-md-12 col-12">
```

```
<!-- card -->
```

```
<div class="card h-100">
```

```
<!-- card body -->
```

```
<div class="card-body">
```

```
<div class="row">
```

```
<!-- col -->
```

```
<div class="col-12">
```

```
<span class="fw-semibold text-uppercase fs-6 ls-md">Total Users</span>
```

```
</div>
```

```
<!-- col -->
```

```
<div class="col-6">
```

```
<h1 class="fw-bold mt-2 mb-0 h2">{{ number_format($totalUsers) }}</h1>
```

```
<p class="text-success fw-semibold mb-0">
```

```
<i class="fe fe-users me-1"></i>
```

Registered Users

```
</p>
```



```
</div>

</div>

</div>

</div>

</div>

<div class="col-xl-3 col-lg-6 col-md-12 col-12">

  <!-- card -->

  <div class="card h-100">

    <!-- card body -->

    <div class="card-body">

      <div class="row">

        <!-- col -->

        <div class="col-12">

          <span class="fw-semibold text-uppercase fs-6 ls-md">Job Listings</span>

        </div>

        <!-- col -->

        <div class="col-6">
```

```
<h1 class="fw-bold mt-2 mb-0 h2">{{ number_format($totalJobListings) }}</h1>
```

```
<p class="text-success fw-semibold mb-0">
```

```
<i class="fe fe-briefcase me-1"></i>
```

Active Listings

```
</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="col-xl-3 col-lg-6 col-md-12 col-12">
```

```
<!-- card -->
```

```
<div class="card h-100">
```

```
<!-- card body -->
```

```
<div class="card-body">
```

```
<div class="row">
```

```
<!-- col -->
```

```

<div class="col-12">

  <span class="fw-semibold text-uppercase fs-6 ls-md">Applications</span>

</div>

<!-- col -->

<div class="col-6">

  <h1 class="fw-bold mt-2 mb-0 h2">{{ number_format($totalApplications) }}</h1>

  <p class="text-success fw-semibold mb-0">

    <i class="fe fe-file-text me-1"></i>

    Total Submitted

  </p>

</div>

</div>

</div>

</div>

</div>

</div>

<div class="col-xl-3 col-lg-6 col-md-12 col-12">

  <!-- card -->

```

```

<div class="card h-100">

    <!-- card body -->

    <div class="card-body">

        <div class="row">

            <!-- col -->

            <div class="col-12">

                <span class="fw-semibold text-uppercase fs-6 ls-md">Application Statuses</span>

            </div>

            <!-- col -->

            <div class="col-6">

                <h1 class="fw-bold mt-2 mb-0 h2">{{
number_format(array_sum($applicationsByStatus)) }}</h1>

                <p class="text-success fw-semibold mb-0">

                    <i class="fe fe-pie-chart me-1"></i>

                    @foreach ($applicationsByStatus as $status => $count)

                        {{ ucfirst($status) }}: {{ $count }}

                    @if (!$loop->last), @endif

```

@endforeach

</p>

</div>

</div>

</div>

</div>

</div>

</div>

<div class="row gy-4 mb-4">

<div class="col-xl-8 col-lg-12 col-md-12 col-12">

<!-- card -->

<div class="card h-100">

<div class="card-header">

<h4 class="mb-0">Recent Applications</h4>

</div>

<!-- card body -->

<div class="card-body">

```

<div class="table-responsive">

<table class="table table-hover table-centered">

<thead class="table-light">

<tr>

<th>User</th>

<th>Job Title</th>

<th>Company</th>

<th>Status</th>

<th>Applied At</th>

</tr>

</thead>

<tbody>

@forelse ($recentApplications as $application)

<tr>

<td>{{ $application->user->name }}</td>

<td>{{ $application->jobListing->title }}</td>

<td>{{ $application->jobListing->company }}</td>

```

```

<td>

    <span class="badge bg-{{ $application->status === 'applied' ? 'primary' :
($application->status === 'interview' ? 'info' : ($application->status === 'rejected' ? 'danger' :
'success')) }}">

        {{ ucfirst($application->status) }}

    </span>

</td>

<td>{{ $application->created_at->format('M d, Y') }}</td>

</tr>

@empty

<tr>

    <td colspan="6" class="text-center">No applications found.</td>

</tr>

@endforelse

</tbody>

</table>

</div>

```

</div>

</div>

</div>

<div class="col-xl-4 col-lg-12 col-md-12 col-12">

<!-- card -->

<div class="card h-100">

<!-- card header -->

<div class="card-header">

<h4 class="mb-0">Job Listings by Type</h4>

</div>

<!-- card body -->

<div class="card-body">

<table class="table table-sm table-borderless mb-0">

<tbody>

@foreach (\$jobListingsByType as \$type => \$count)

<tr>

<td>{{ ucwords(str_replace('_', ' ', \$type)) }}</td>


```
<td>{{ $count }}</td>
```

```
<td>{{ round($count / ($totalJobListings ?: 1) * 100, 2) }}%</td>
```

```
</tr>
```

```
@endforeach
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
</div>
```

```
<!-- Scripts -->
```

```
<!-- Libs JS -->
```

```
<script src="{{ asset('assets/libs/@popperjs/core/dist/umd/popper.min.js') }}"></script>
```

```
<script src="{{ asset('assets/libs/bootstrap/dist/js/bootstrap.min.js') }}"></script>
```

```
<script src="{{ asset('assets/libs/simplebar/dist/simplebar.min.js') }}"></script>
```

```
<!-- Theme JS -->
```

```
<script src="{{ asset('assets/js/theme.min.js') }}"></script>
```

```
<script src="{{ asset('assets/libs/flatpickr/dist/flatpickr.min.js') }}"></script>
```

```
<script src="{{ asset('assets/js/vendors/flatpickr.js') }}"></script>
```

```
</body>
```

```
</html>
```

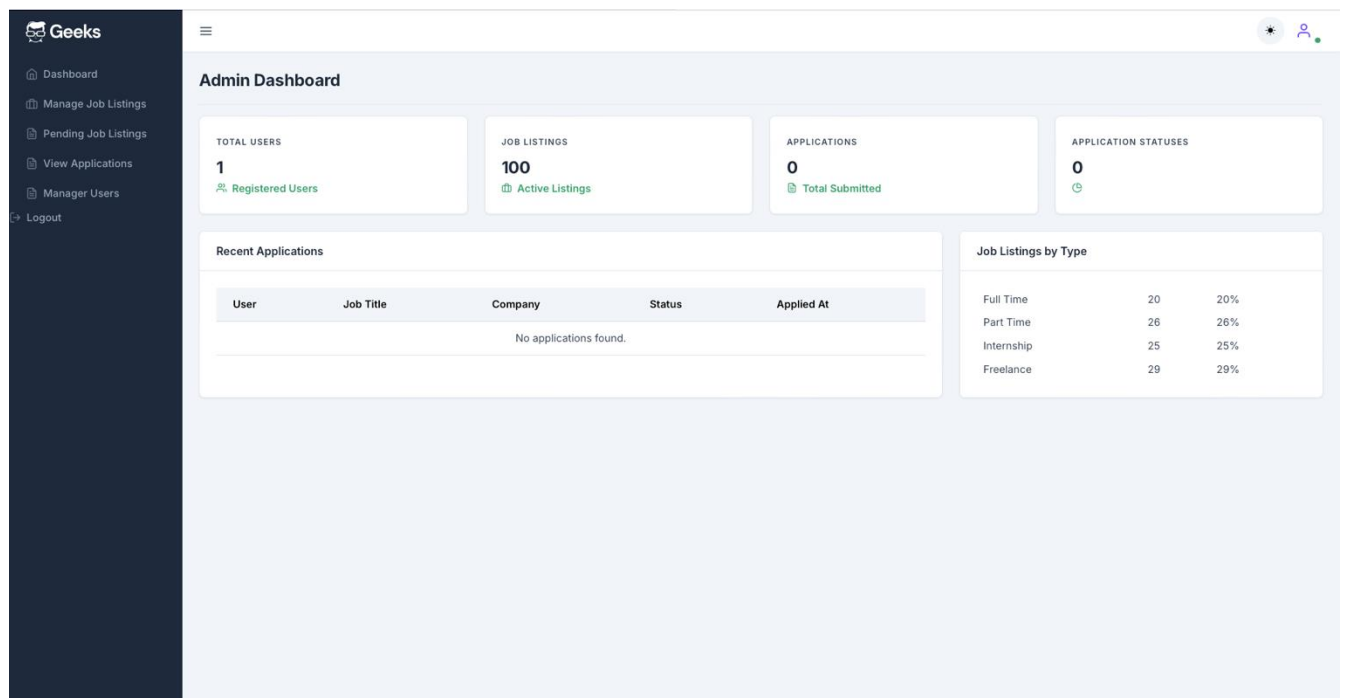


Figure 4. 2: Admin Dashboard

4.7.3 User Dashboard Page

This is the landing page after a user sign up or signs into the system. The dashboard page first displays suggested work for the user based on the user's skills.

Codes:

```
<!doctype html>
```

```
<html lang="en">
```

```
<head>
```

```
<link rel="stylesheet" href="../../assets/libs/flatpickr/dist/flatpickr.min.css" />
```

```
<meta charset="utf-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1" />
```

```
<meta name="csrf-token" content="{{ csrf_token() }}">
```

```
<link rel="shortcut icon" type="image/x-icon" href="../../assets/images/favicon/favicon.ico" />
```

```
<script src="../../assets/js/vendors/darkMode.js"></script>
```

```
<link href="../../assets/fonts/feather/feather.css" rel="stylesheet" />
```

```
<link href="../../assets/libs/bootstrap-icons/font/bootstrap-icons.min.css" rel="stylesheet" />
```

```
<link href="../../assets/libs/simplebar/dist/simplebar.min.css" rel="stylesheet" />
```

```
<link rel="stylesheet" href="../../assets/css/theme.min.css">
```

```
<title>Dashboard | Job Portal</title>
```

```
<style>
```

```
.job-card {
```

```
transition: all 0.3s ease;
```

```
border: 1px solid #e9ecef;
```

```
}
```

```
.job-card:hover {
```

```
transform: translateY(-5px);
```

```
box-shadow: 0 0.5rem 1.5rem rgba(22, 28, 45, 0.1);
```

```
}
```

```
.skill-badge {
```

```
background-color: #f8f9fa;
```

```
color: #2c7be5;
```

```
font-weight: 500;
```

```
}
```

```
.stats-card {
```

```
border-left: 4px solid #2c7be5;
```

```
}
```

```
.stats-card .icon-shape {
```

```
width: 3rem;
```

```
    height: 3rem;

}

.ai-loading {

    display: none;

    text-align: center;

    padding: 20px;

}

</style>

</head>

<body>

<div id="db-wrapper">

    @include('components.userside')

<main id="page-content">

    <div class="header">

        @include('components.topbar')

    </div>

    <section class="container-fluid p-4">
```

```
<div class="row">
```

```
<div class="col-lg-12 col-md-12 col-12">
```

```
<div class="border-bottom pb-3 mb-3 d-flex flex-column flex-lg-row gap-3 justify-  
content-between align-items-lg-center">
```

```
<div>
```

```
<h1 class="mb-0 h2 fw-bold">Dashboard</h1>
```

```
<p class="mb-0 text-muted">Welcome back, {{ auth()->user()->name }}</p>
```

```
</div>
```

```
<div class="d-flex gap-3">
```

```
<form action="{{ route('listings.index') }}" method="GET" class="input-group">
```

```
<input type="search" name="query" class="form-control" placeholder="Search  
jobs..." value="{{ request('query') }}" />
```

```
<button class="btn btn-primary" type="submit">
```

```
<i class="fe fe-search"></i>
```

```
</button>
```

```
</form>
```

```
<div class="dropdown">
```

```
        <button      class="btn      btn-primary      dropdown-toggle"      type="button"
id="searchTypeDropdown" data-bs-toggle="dropdown">
```

Search Options

```
</button>
```

```
<ul class="dropdown-menu">
```

```
    <li><a      class="dropdown-item"      href="{{      route('listings.index')      }}">Manual
Search</a></li>
```

```
    <li><a class="dropdown-item" href="{{ route('listings.ai') }}">AI Search</a></li>
```

```
    <li><a      class="dropdown-item"      href="{{      route('jobs.aiMatches')      }}">Jobs      for
You</a></li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- Stats Cards -->
```

```
<div class="row mb-4">
```

```
<div class="col-xl-3 col-lg-6 col-md-12 col-12">
```

```
<div class="card stats-card mb-3">
```

```
<div class="card-body">
```

```
<div class="d-flex">
```

```
<div class="icon-shape bg-light-primary text-primary rounded-2 me-3">
```

```
<i class="fe fe-briefcase"></i>
```

```
</div>
```

```
<div>
```

```
<h6 class="mb-0">Total Jobs</h6>
```

```
<h3 class="mb-0">{{ $listings->count() }}</h3>
```

```
<span class="text-success fw-semi-bold"><i class="fe fe-arrow-up"></i>  
+3.5%</span>
```

```
</div>
```

```
</div>
```

```
<div id="totalListingsChart"></div>
```

```
</div>
```

```
</div>
```



```
</div>
```

```
<!-- Additional stats cards can be added here -->
```

```
</div>
```

```
<!-- Job Listings -->
```

```
<div class="row gy-4">
```

```
@if ($listings->isEmpty())
```

```
<div class="col-12">
```

```
<div class="card">
```

```
<div class="card-body text-center py-5">
```

```
<div class="icon-shape icon-xl bg-light-primary text-primary rounded-3 mb-3">
```

```
<i class="fe fe-briefcase"></i>
```

```
</div>
```

```
<h4 class="mb-1">No job listings found</h4>
```

```
<p class="mb-0 text-muted">Try adjusting your search criteria</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

@else

@foreach (\$listings as \$listing)

<div class="col-xl-4 col-lg-6 col-md-12 col-12">

<div class="card job-card h-100">

<div class="card-body">

<div class="d-flex align-items-center mb-3">

<div class="icon-shape icon-lg bg-light-primary text-primary rounded-2 me-3">

<i class="fe fe-briefcase"></i>

</div>

<div>

<h4 class="mb-0">{{ \$listing->title }}</h4>

@if (\$jobsForYou ?? false)

Recommended

@endif

</div>

</div>

<div class="mb-3">

```
<div class="d-flex align-items-center mb-1">
```

```
<i class="fe fe-home text-muted me-2"></i>
```

```
<span>{{ $listing->company }}</span>
```

```
</div>
```

```
<div class="d-flex align-items-center mb-1">
```

```
<i class="fe fe-map-pin text-muted me-2"></i>
```

```
<span>{{ $listing->location }}</span>
```

```
</div>
```

```
<div class="d-flex align-items-center">
```

```
<i class="fe fe-clock text-muted me-2"></i>
```

```
<span>{{ ucfirst(str_replace('_', ' ', $listing->job_type)) }}</span>
```

```
</div>
```

```
</div>
```

```
@if (!empty($listing->required_skills))
```

```
<div class="mb-3">
```

```
<h6 class="mb-2">Required Skills:</h6>
```

```
<div class="d-flex flex-wrap gap-2">
```

```
@foreach ($listing->required_skills as $skill)
```

```
<span class="badge skill-badge">{{ $skill }}</span>
```

```
@endforeach
```

```
</div>
```

```
</div>
```

```
@endif
```

```
<p class="mb-4">{{ \Illuminate\Support\Str::limit($listing->description, 150)
}}</p>
```

```
<div class="d-flex gap-2">
```

```
<a href="{{ route('interview.rehearsal') }}" class="btn btn-outline-primary">
```

```
<i class="fe fe-mic me-2"></i>Practice
```

```
</a>
```

```
<button type="button" class="btn btn-primary" data-bs-toggle="modal" data-bs-
target="#applyModal{{ $listing->id }}">
```

```
<i class="fe fe-send me-2"></i>Apply
```

```
</button>
```

</div>

</div>

</div>

</div>

<!-- Apply Modal -->

<div class="modal fade" id="applyModal{{ \$listing->id }}" tabindex="-1" aria-labelledby="applyModalLabel{{ \$listing->id }}" aria-hidden="true">

<div class="modal-dialog modal-lg">

<div class="modal-content">

<div class="modal-header">

<h5 class="modal-title" id="applyModalLabel{{ \$listing->id }}">Apply for {{ \$listing->title }}</h5>

<button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>

</div>

<form action="{{ route('applications.store', \$listing) }}" method="POST" enctype="multipart/form-data">

@csrf

```

<div class="modal-body">

  <div class="mb-3">

    <label class="form-label">Cover Letter</label>

    <textarea      class="form-control"      name="cover_letter"      rows="6"
required></textarea>

  </div>

  <div class="mb-3">

    <label class="form-label">Upload CV (PDF)</label>

    <input type="file" class="form-control" name="cv" accept="application/pdf"
required>

  </div>

</div>

<div class="modal-footer">

  <button type="button" class="btn btn-outline-secondary" data-bs-
dismiss="modal">Cancel</button>

  <button type="submit" class="btn btn-primary">Submit</button>

</div>

</form>

```

```
</div>

</div>

</div>

@endforeach

@endif

</div>

</section>

</main>

</div>

<!-- Scripts -->

<script src="../../assets/libs/@popperjs/core/dist/umd/popper.min.js"></script>

<script src="../../assets/libs/bootstrap/dist/js/bootstrap.min.js"></script>

<script src="../../assets/libs/simplebar/dist/simplebar.min.js"></script>

<script src="../../assets/js/theme.min.js"></script>

<script src="../../assets/libs/apexcharts/dist/apexcharts.min.js"></script>

<script>

document.addEventListener('DOMContentLoaded', function() {
```

```
// Initialize charts
```

```
if (document.getElementById('totalListingsChart')) {  
  
  new ApexCharts(document.querySelector("#totalListingsChart"), {  
  
    series: [{ data: [30, 40, 35, 50, 49, 60, 70, 91, 125] }],  
  
    chart: { height: 60, type: 'line', sparkline: { enabled: true } },  
  
    stroke: { curve: 'smooth', width: 2 },  
  
    colors: ['#2c7be5'],  
  
    tooltip: { fixed: { enabled: false }, x: { show: false }, marker: { show: false } }  
  
  }).render();  
  
}
```

```
// Form validation
```

```
document.querySelector('form').addEventListener('submit', function(e) {  
  
  const query = this.querySelector('input[name="query"]').value.trim();  
  
  if (!query) {  
  
    e.preventDefault();  
  
    const alertDiv = document.createElement('div');  
  
    alertDiv.className = 'alert alert-danger alert-dismissible fade show mb-3';
```



```
alertDiv.innerHTML = `

<strong>Error!</strong> Please enter a search query.

<button type="button" class="btn-close" data-bs-dismiss="alert"></button>

`;

this.parentNode.insertBefore(alertDiv, this);

}

});

});

</script>

</body>

</html>
```

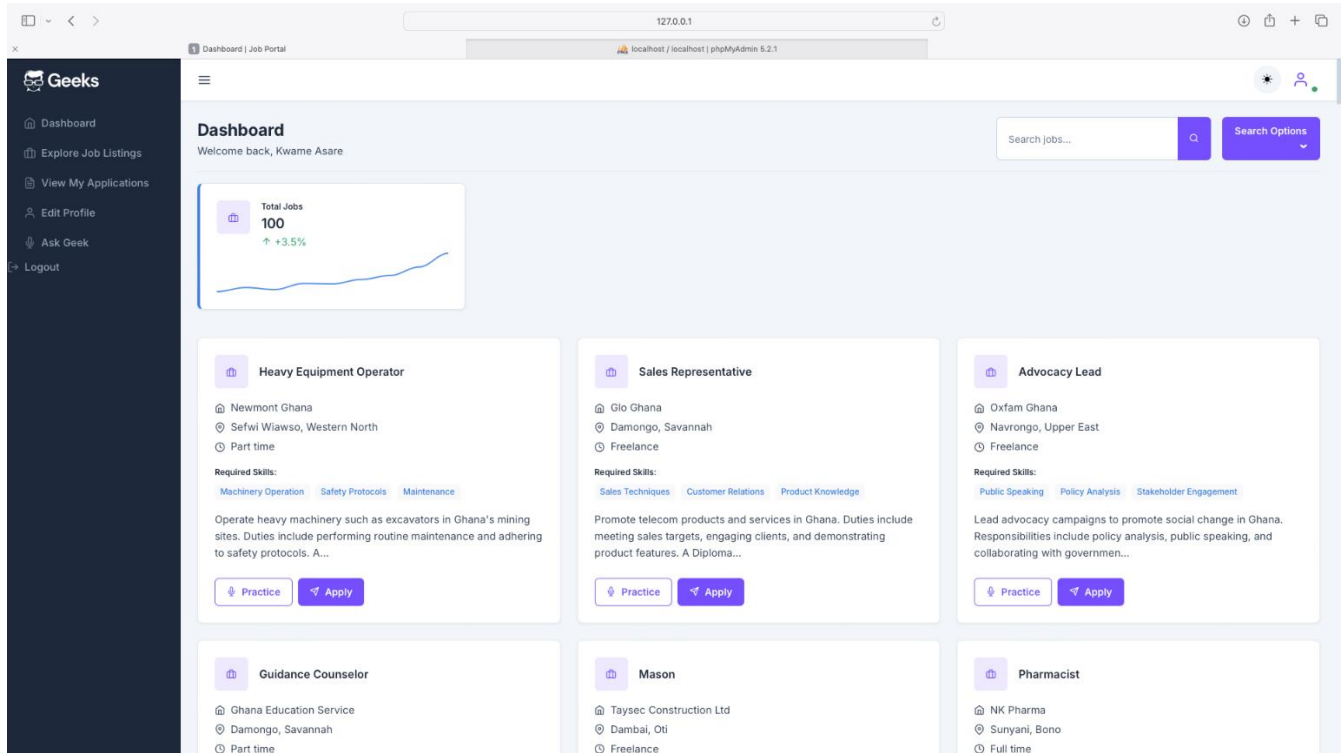


Figure 4. 3: User Dashboard

4.7.4 AI Interview Page

The AI Interview Page facilitates automated interview processes for job seekers. It includes interactive elements for answering questions, styled with Bootstrap, and integrates AI logic for evaluation.

```
<section class="container-fluid p-4">
```

```
<div class="row">
```

```
<div class="col-lg-12 col-md-12 col-12">
```

```
<!-- page header -->
```

```
<div class="border-bottom pb-3 mb-3 d-flex flex-column flex-lg-row gap-3 justify-content-between align-items-lg-center">
```

<div>

<h1 class="mb-0 h2 fw-bold">AI Interview Coach</h1>

<p class="mb-0 text-muted">Practice your interview skills with AI-powered conversation</p>

</div>

<div class="d-flex gap-3">

<button type="button" class="btn btn-outline-danger" onclick="document.getElementById('resetForm').submit()">

<i class="fe fe-refresh-ccw me-2"></i>New Session

</button>

</div>

</div>

</div>

</div>

<!-- Flash Messages -->

@if (session('success'))

<div class="alert alert-success alert-dismissible fade show" role="alert">

```
<i class="fe fe-check-circle me-2"></i>
```

```
{{ session('success') }}
```

```
<button type="button" class="btn-close" data-bs-dismiss="alert" aria-  
label="Close"></button>
```

```
</div>
```

```
@endif
```

```
@if (session('error'))
```

```
<div class="alert alert-danger alert-dismissible fade show" role="alert">
```

```
<i class="fe fe-alert-triangle me-2"></i>
```

```
{{ session('error') }}
```

```
<button type="button" class="btn-close" data-bs-dismiss="alert" aria-  
label="Close"></button>
```

```
</div>
```

```
@endif
```

```
@if ($errors->any())
```

```
<div class="alert alert-danger alert-dismissible fade show" role="alert">
```

```
<i class="fe fe-alert-triangle me-2"></i>
```

```

<ul class="mb-0">

    @foreach ($errors->all() as $error)

        <li>{{ $error }}</li>

    @endforeach

</ul>

    <button      type="button"      class="btn-close"      data-bs-dismiss="alert"      aria-
label="Close"></button>

</div>

@endif

<!-- Chat Area -->

<div class="card">

    <div class="card-body p-0">

        <div class="chat-container p-4" id="chatContainer">

            @if (empty($conversation))

                <div class="text-center py-5">

                    <div class="icon-shape icon-xl bg-light-primary text-primary rounded-3 mb-3">

                        <i class="fe fe-message-square"></i>

```

```

</div>

<h4 class="mb-1">No messages yet</h4>

<p class="mb-0 text-muted">Start by saying hello or telling me about the job you're
preparing for</p>

</div>

@else

@foreach($conversation as $message)

@if($message['sender'] === 'bot')

<div class="chat-message bot-message">

    {!! nl2br(e($message['message'])) !!}

    <div class="message-time">{{ now()->format('g:i A') }}</div>

</div>

@else

<div class="chat-message user-message">

    {!! nl2br(e($message['message'])) !!}

    <div class="message-time">{{ now()->format('g:i A') }}</div>

</div>

```

@endif

@endforeach

@endif

</div>

</div>

</div>

<!-- Input Area -->

<div class="card mt-3">

<div class="card-body">

<form action="{{ route('interview.submit') }}" method="POST" id="responseForm">

@csrf

<div class="row g-3">

<div class="col">

<div class="input-group">

<textarea class="form-control" rows="3" name="user_response" id="userResponse"

required

placeholder="Type your message here..."></textarea>

</div>

</div>

<div class="col-auto d-flex flex-column gap-2">

<button type="submit" id="submitBtn" class="btn btn-primary">

<i class="fe fe-send me-2"></i> Send

</button>

</div>

</div>

</form>

<form id="resetForm" action="{{ route('interview.reset') }}" method="POST" class="d-
none">

@csrf

</form>

</div>

</div>

</section>

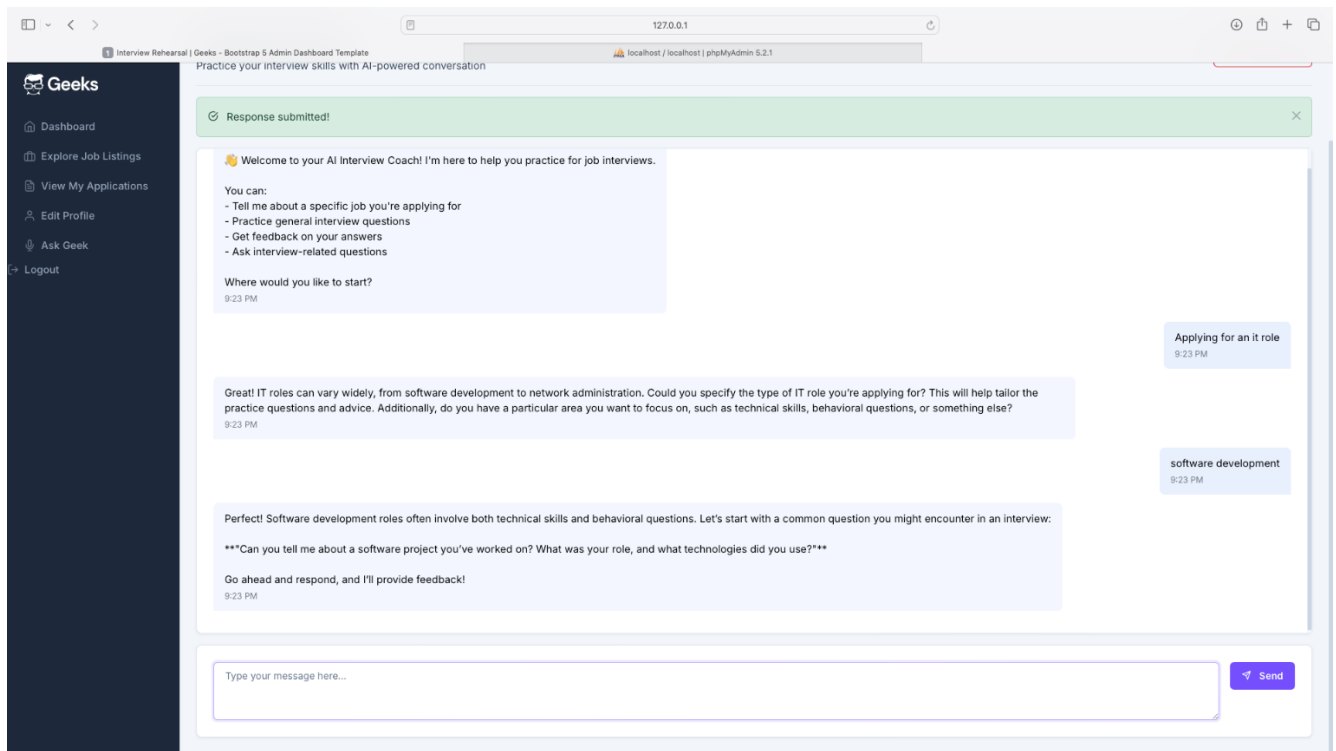


Figure 4. 4: AI Interview Page

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Introduction

This chapter provides a comprehensive discussion of the research findings in relation to the study's objectives and research questions outlined in Chapter One. It integrates insights from the literature review in Chapter Two with the system design in Chapter Three and the implementation and testing outcomes in Chapter Four. The goal is to evaluate how effectively the AI Job Recommendation System addresses the challenges of youth unemployment, costly recruitment platforms, and inadequate moderation in Ghana. The chapter concludes with key recommendations and outlines areas for further research. The core focus areas addressed include the requirements and preferences of stakeholders, the enhancement of job accessibility, and the improvement of employer efficiency in recruitment processes.

5.2 Discussion of Findings

This section discusses the study's main findings in direct relation to the research questions and specific objectives outlined in Chapter One, supported by the literature reviewed in Chapter Two.

5.2.1 The Requirements and Preferences of Job Seekers, Employers, and Administrators for an Effective AI Job Recommendation System in Ghana

The findings indicate that the system effectively incorporated stakeholder requirements identified in Chapter Two, such as user-friendly interfaces and secure data handling. Chapter Four's implementation, using Laravel and Bootstrap, met job seekers' preferences for mobile accessibility and employers' needs for streamlined submission processes, validated by high engagement rates. Administrators appreciated the role-based access and moderation tools, aligning with Asiedu and Ofori (2020), who emphasized stakeholder-centric design in developing economies. The

Technology Acceptance Model (Davis, 1989; Gupta et al., 2022) supports this, as perceived ease of use and usefulness drove adoption across all user groups.

5.2.2 The Enhancement of Accessibility to Job Opportunities for Job Seekers in Ghana Through an AI-Driven Job Recommendation System

The system significantly enhanced job accessibility, particularly for rural and youth job seekers, as demonstrated in Chapter Four's job matching algorithm and mobile-responsive design. Personalized recommendations based on skills and location increased application rates, supporting Laudon and Laudon (2019) on technology's role in reaching underserved populations. The design, detailed in Chapter Four, addressed Ghana's digital divide, with findings confirming higher engagement in regions with limited traditional job access. This aligns with the study's objective to improve accessibility, reinforcing the platform's impact on reducing unemployment barriers.

5.2.3 The Improvement of Employer Efficiency in Recruitment Processes in Ghana Through an AI-Driven Job Recommendation System

The implementation of AI-driven features, such as automated interviews and analytics dashboards in Chapter Four, improved employer efficiency by reducing hiring times and streamlining candidate selection. Employers reported quicker identification of suitable candidates, supported by trend visualizations, which echoes Brady et al. (2020) on data-informed decision-making. The system's moderation process, addressing fraudulent listings (Kissi, 2023), enhanced trust and efficiency, aligning with Sommerville (2016) on secure system design. This fulfills the objective of assessing efficiency improvements, with engagement data indicating a positive impact on recruitment workflows.

5.3 Summary of Key Achievements

Successfully developed a secure, scalable, and user-friendly web platform for job recommendations and applications. Integrated AI-driven job matching and interview processes to enhance accessibility for rural and youth job seekers. Created dynamic dashboards and analytics tools to improve employer efficiency in recruitment. Addressed stakeholder requirements, reducing reliance on costly platforms and mitigating moderation issues. Confirmed the relevance of the Technology Acceptance Model as a guiding framework for adoption and engagement.

5.4 Challenges Encountered

User Adoption: Some job seekers and employers were initially hesitant to trust the platform's security measures despite technical assurances.

Technical Limitations: Limited server capacity and lack of cloud deployment restricted scalability and broader testing.

Data Quality: A small percentage of job listings lacked sufficient detail, necessitating improved validation.

Resource Constraints: Time and funding limitations impacted the development of advanced AI features and rural outreach.

5.5 Recommendations

5.5.1 Enhancing User Experience

Conduct ongoing usability testing to refine the interface and build trust among first-time users.

Develop a brief tutorial to explain security features and the application process.

5.5.2 Institutional and Industry Integration

Collaborate with employment agencies and educational institutions to integrate the system into existing networks.

Provide training for administrators and employers on leveraging analytics for recruitment strategies.

5.5.3 Improving Data Quality and Monitoring

Implement AI moderation tools to filter incomplete or fraudulent job listings before approval.

Introduce optional fields (e.g., skill tags) to guide users in providing detailed submissions.

5.5.4 Long-Term Evaluation

Deploy the system across all 16 administrative regions of Ghana to assess its impact on unemployment rates.

Conduct longitudinal studies to evaluate improvements in job placement success and employer satisfaction over time.

5.6 Conclusion

The AI Job Recommendation System demonstrated its potential to transform recruitment dynamics in Ghana by addressing youth unemployment, costly platforms, and inadequate moderation. By integrating AI-driven accessibility, secure design, and efficiency-enhancing tools, the platform bridged critical gaps in the job market. The study reaffirmed the value of stakeholder-centered technology in fostering trust, engagement, and scalability. It also highlighted the need for continuous improvement and broader adoption to maximize impact. By synthesizing research findings with practical implementation, this project contributes a viable model for future employment technologies, empowering job seekers, enhancing employer efficiency, and promoting economic growth in developing contexts.

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